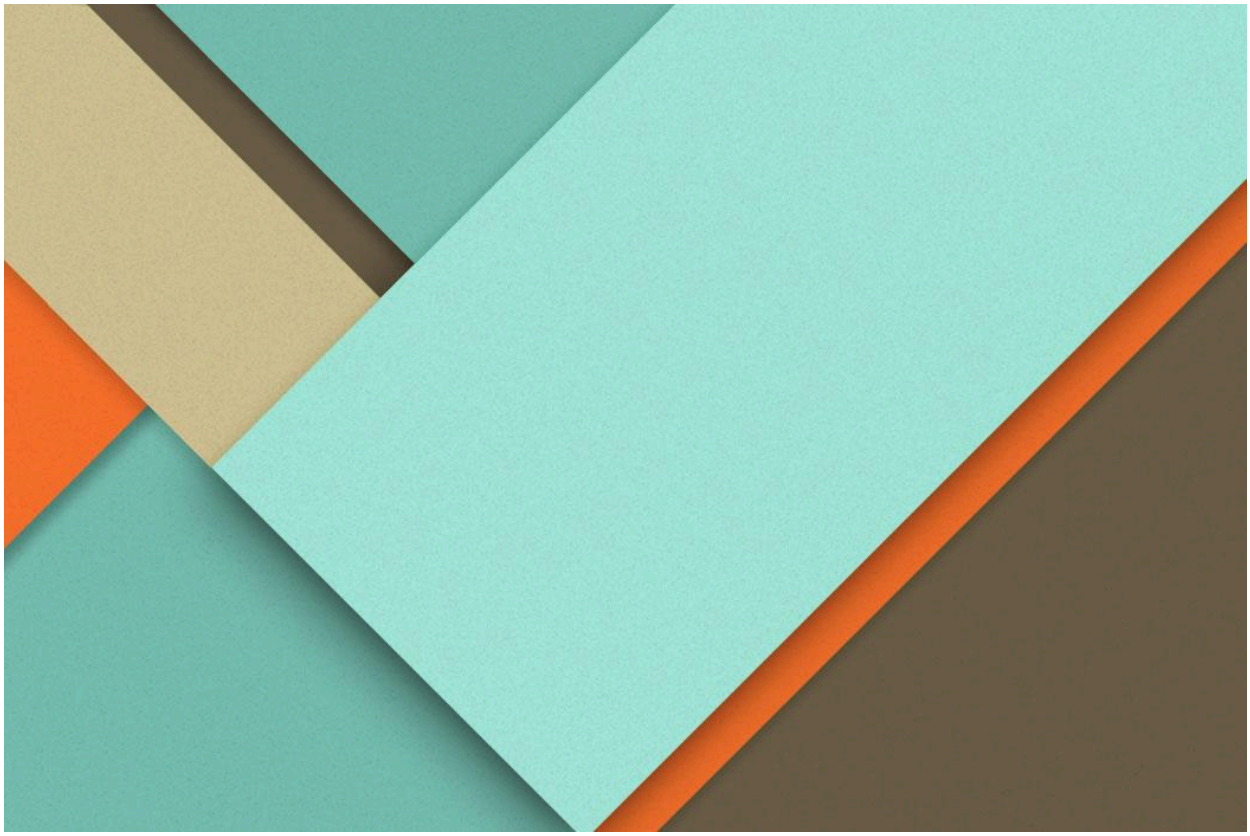


# Career Roadmap

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Software Engineer

## Short-Term Goal (6–12 Months)

My short-term goal is to build a strong foundation in programming while beginning to develop skills in computational biology and bioinformatics. I aim to gain practical experience with tools and techniques used in computational drug discovery and medical data analysis, while building a portfolio that reflects both my coding ability and scientific interests

## Skills to Develop

- Molecular docking (predicting how molecules interact with biological targets)
- Virtual screening (analysing large datasets of compounds computationally)
- Using UCSF ChimeraX for molecular visualization and analysis
- Basic understanding in medical imaging data
- Scientific blogging to document learning and explain concepts clearly
- Academic writing skills for publishing research papers
- Advanced Python Data Science techniques
- Advanced deep Java programming knowledge and expertise
- Become proficient in Blockchain deployment & Advanced Security Techniques

## Experience Needed:

- Working on beginner bioinformatics or data-related projects
- Practicing molecular visualization using ChimeraX
- Exploring basic medical imaging datasets
- Reading and summarising scientific papers
- Writing blog posts about what I learn

## Support or Mentorship Required:

- Guidance from lecturers, professors and experienced developers
- Read books in Biology, Pharmacology, Pathology, Biochemistry, Physiology & Bioinformatics while practicing with software tools for medical data analysis and predictions
- Peer collaboration and group learning
- Access to open-source research and datasets

## Long-Term Goal (3–5+ Years):

My long-term goal is to pursue a PhD in computational biology or bioinformatics and contribute to research in computational drug discovery and medical data analysis. I want to work at the intersection of software development, biology, and healthcare, using computational methods to solve complex scientific and medical problems.

## Skills to Develop

- Advanced molecular docking and simulation techniques
- Large-scale virtual screening and data analysis
- Advanced use of tools like UCSF ChimeraX
- Computational analysis of medical imaging processing and interpretation
- Scientific writing and publishing in academic journals
- Technical communication through blogging and research sharing

## Experience Needed:

- Research experience in computational biology or related fields
- Exposure to projects involving biological and medical datasets
- Working with imaging data or healthcare-related systems
- Writing and contributing to research papers
- Attending workshops, seminars, or conferences

## Support or Mentorship Required:

- Academic mentors in bioinformatics or computational science
- University programs aligned with computational biology
- Collaboration with research labs or healthcare-related projects
- Access to scientific tools, datasets, and research communities